

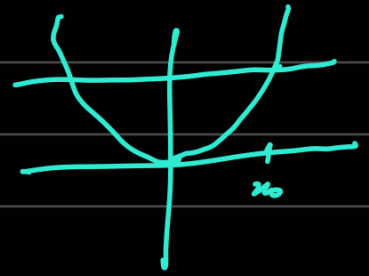
1. Period of oscillation: Determine the period of oscillation as a function of the energy, say,  $E$ , when a particle of mass  $m$  is moving in one dimension (along the  $x$ -axis) under the influence of the following potentials: (a)  $U = A|x|^{2n}$  (with  $A > 0$  and  $n > 0$ ), (b)  $U(x) = -U_0/\cosh^2(\alpha x)$  (for  $-U_0 < E < 0$ ), and (c)  $U(x) = U_0 \tan^2(\alpha x)$ .

a)

$$\frac{T}{4} \int dt = \sqrt{\frac{m}{2}} \int_0^{x_0} \frac{dx}{\sqrt{E - Ax^{2n}}}$$

$$\frac{T}{4} = \sqrt{\frac{m}{2}} \int_0^{x_0} \frac{dx}{\sqrt{E - Ax^{2n}}}$$

$$T = \sqrt{\frac{8m}{E}} \int_0^{\left(\frac{E}{A}\right)^{\frac{1}{2n}}} \frac{dx}{\sqrt{1 - \left(\frac{x}{x_0}\right)^{2n}}}$$



$$E = Ax_0^{2n}$$

$$x_0 = \left(\frac{E}{A}\right)^{\frac{1}{2n}}$$

$$x = sx_0$$

$$s = \frac{x}{x_0}$$

$$T = x_0 \sqrt{\frac{8m}{E}} \int_0^1 \frac{ds}{\sqrt{1 - s^{2n}}}$$

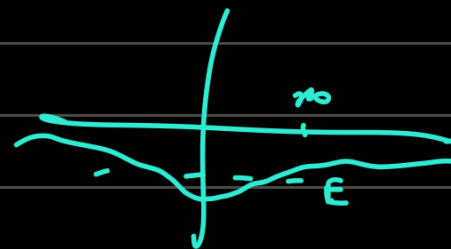
$$T = C \left(\frac{E}{A}\right)^{\frac{1}{2n}} \sqrt{\frac{8m}{E}}$$

b)  $U = \frac{-U_0}{\cosh^2 \alpha x}$

$$\frac{T}{4} = \sqrt{\frac{m}{2}} \int_0^{x_0} \frac{dx}{\sqrt{E + \frac{U_0}{\cosh^2 \alpha x}}}$$

$$T = \sqrt{8m} \int_0^{x_0} \frac{\cosh \alpha x \, dx}{\sqrt{E \cosh^2 \alpha x + U_0}}$$

$$T = \sqrt{\frac{8m}{\alpha^2}} \int_0^{x_0} \frac{\alpha \cosh \alpha x \, dx}{\sqrt{-\epsilon \cosh^2 \alpha x + U_0}}$$



let  $E = -\epsilon$

$$\epsilon = \frac{U_0}{\cosh^2 \alpha x}$$

$$T = \sqrt{\frac{8m}{\alpha^2}} \int_0^{x_0} \frac{\alpha \cosh \alpha x \, dx}{\sqrt{U_0 - \epsilon - \epsilon \sinh^2 \alpha x}} \quad u_0^2 = \frac{U_0}{\epsilon} - 1$$

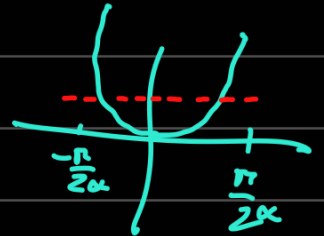
$$T = \sqrt{\frac{8m}{\alpha^2 \epsilon}} \int_0^{u_0} \frac{du}{\sqrt{\left(\frac{U_0 - \epsilon}{\epsilon}\right) - u^2}} = \sqrt{\frac{8m}{\alpha^2 \epsilon}} \int_0^{u_0} \frac{du}{\sqrt{u_0^2 - u^2}}$$

$$T = \frac{1}{u_0} \sqrt{\frac{8m}{\alpha^2 \epsilon}} \int_0^{u_0} \frac{du}{\sqrt{1 - \frac{u^2}{u_0^2}}} \quad u = su_0$$

$$T = \sqrt{\frac{8m}{\alpha^2 \epsilon}} \int_0^1 \frac{ds}{\sqrt{1 - s^2}}$$

$$T = \frac{\pi}{\alpha} \sqrt{\frac{2m}{|\epsilon|}}$$

c)  $V(x) = U_0 \tan^2 \alpha x$



$$E = \frac{1}{2} m \dot{x}^2 + U_0 \tan^2 \alpha x \quad \sqrt{\frac{2}{m}} \sqrt{E - U_0 \tan^2 \alpha x} = \frac{dx}{dt}$$

$$\frac{T}{2} = \sqrt{\frac{m}{2}} \int_{-x_0}^{x_0} \frac{dx}{\sqrt{E - U_0 \tan^2 \alpha x}} = \sqrt{\frac{m}{2U_0}} \int_{-x_0}^{x_0} \frac{dx}{\sqrt{\tan^2 \alpha x_0 - \tan^2 \alpha x}}$$

$$\Rightarrow \frac{T}{2} = \frac{1}{\tan \alpha x_0} \sqrt{\frac{m}{2U_0}} \int_{-x_0}^{x_0} \frac{dx}{\sqrt{1 - \frac{\tan^2 \alpha x}{\tan^2 \alpha x_0}}}$$

$E = U_0 \tan^2 \alpha x_0$

$$\sin \epsilon = \frac{\tan \alpha x}{\tan \alpha x_0} \Rightarrow \cos \epsilon \, d\epsilon = \frac{\alpha \sec^2 \alpha x}{\tan \alpha x_0} \, dx$$

$$\frac{T}{2} \propto \sqrt{\frac{m}{2V_0}} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{d\varepsilon}{\sec^2 \alpha \chi}$$

and  $\sec^2 \alpha \chi = 1 + \tan^2 \alpha \chi_0 \cdot \sin^2 \varepsilon$

$$\frac{T}{2} \propto \sqrt{\frac{m}{2V_0}} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{d\varepsilon}{1 + K^2 \sin^2 \varepsilon}$$

$K = \tan \alpha \chi_0$

$$\frac{T}{2} \propto \sqrt{\frac{m}{2V_0}} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\sec^2 \varepsilon d\varepsilon}{\sec^2 \varepsilon + K^2 \tan^2 \varepsilon}$$

$$\frac{T}{2} \propto \sqrt{\frac{m}{2V_0}} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\sec^2 \varepsilon d\varepsilon}{1 + (K^2 + 1) \tan^2 \varepsilon}$$

$\tan \varepsilon = p$

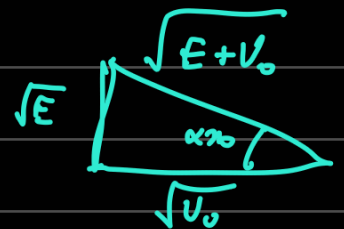
$\sec^2 \varepsilon d\varepsilon = dp$

$$\frac{T}{2} \propto \sqrt{\frac{m}{2V_0}} \int_{-\infty}^{\infty} \frac{dp}{1 + (K^2 + 1)p^2} = \frac{1}{K^2 + 1} \int_{-\infty}^{\infty} \frac{dp}{\left(\frac{1}{K^2 + 1}\right) + p^2}$$

$$\frac{T}{2} \propto \sqrt{\frac{m}{2V_0}} \frac{1}{(K^2 + 1)^{1/2}} \pi$$

$$\frac{T}{2} \propto \frac{\pi}{\sqrt{2V_0}} \cos \alpha \chi_0$$

and  $E = V_0 \tan^2 \alpha \chi_0$



$$\frac{T}{2} \propto \sqrt{\frac{m}{2V_0}} \frac{\sqrt{V_0}}{\sqrt{E + V_0}}$$

$$\boxed{T = \frac{\pi}{\sqrt{E + V_0}} \sqrt{\frac{2m}{E + V_0}}}$$

2. Explicitly establishing the virial theorem: Evaluate the time averages (i.e. the average over one complete period) of the kinetic and potential energies, say,  $T$  and  $U$ , for a particle that is moving along an elliptic orbit in the Keplerian central potential, and establish the virial theorem for the case, viz. that  $2\langle T \rangle = -\langle U \rangle$ , where the angular brackets denote the averages.

$$t = \sqrt{\frac{ma}{\alpha}} a (\xi - e \sin \xi) \Rightarrow dt = \sqrt{\frac{ma}{\alpha}} a (1 - e \cos \xi) d\xi$$

$$r = a(1 - e \cos \xi) \Rightarrow \frac{dr}{dt} = \frac{dr}{d\xi} \frac{d\xi}{dt} = (e \sin \xi) \left( \sqrt{\frac{\alpha}{ma}} \frac{1}{1 - e \cos \xi} \right)$$

$$U = -\frac{\alpha}{r} = -\frac{\alpha}{a(1 - e \cos \xi)} \Rightarrow \langle U \rangle = \frac{\int_{-2\pi}^{2\pi} -\frac{\alpha}{r} dt}{\int dt}$$

$$\Rightarrow \langle U \rangle = \frac{\int_0^{2\pi} \left( -\frac{\alpha}{a(1 - e \cos \xi)} \right) \sqrt{\frac{ma}{\alpha}} a (1 - e \cos \xi) d\xi}{T_0} \rightarrow \text{time period}$$

$$\Rightarrow \langle U \rangle = -a \sqrt{\frac{m\alpha}{a}} \cdot \frac{2\pi}{T_0} = -\frac{\alpha}{a}$$

$$T = \frac{1}{2} m \dot{r}^2 + \frac{L^2}{2mr^2}$$

$$\langle T \rangle = \frac{1}{T_0} \left( \frac{1}{2} m \int \dot{r}^2 dt + \frac{L^2}{2m} \int \frac{dt}{r^2} \right)$$

$$\langle T \rangle = \frac{1}{T_0} \left( \frac{1}{2} m \int \dot{r} dr + \frac{L^2}{2m} \int \frac{dt}{r^2} \right)$$

$$\text{and } \dot{r} = (e \sin \xi) \left( \sqrt{\frac{\alpha}{ma}} \frac{1}{1 - e \cos \xi} \right)$$

$$dr = ae \sin \xi d\xi$$

$$\langle T \rangle = \frac{1}{T_0} \left( \frac{m e^2}{2} \sqrt{\frac{\alpha a}{m}} \int \frac{\sin^2 \xi}{1 - e \cos \xi} d\xi + \frac{L^2}{2m} \sqrt{\frac{m}{a\alpha}} \int \frac{d\xi}{(1 - e \cos \xi)} \right)$$



$$\text{Now } e^2 = 1 + \frac{2El^2}{m\alpha^2}$$

$$\Rightarrow l^2 = \frac{(e^2 - 1)m\alpha^2}{2E}$$

$$\Rightarrow \frac{l^2}{2m} \sqrt{\frac{m}{a\alpha}} = \frac{(e^2 - 1)\alpha^2}{4E} \sqrt{\frac{m}{a\alpha}}$$

$$\text{and } E = -\frac{\alpha}{2a}$$

$$\begin{aligned} \Rightarrow \frac{l^2}{2m} \sqrt{\frac{m}{a\alpha}} &= -\frac{(e^2 - 1)\alpha a}{2} \sqrt{\frac{m}{a\alpha}} \\ &= -\frac{m(e^2 - 1)}{2} \sqrt{\frac{\alpha a}{m}} \end{aligned}$$

$$\langle T \rangle = \frac{1}{T_0} \left( \frac{mc^2}{2} \sqrt{\frac{\alpha a}{m}} \int \frac{\sin^2 \xi}{1 - e\cos \xi} d\xi - \frac{m(e^2 - 1)}{2} \sqrt{\frac{\alpha a}{m}} \int \frac{d\xi}{(1 - e\cos \xi)} \right)$$

$$\langle T \rangle = \frac{1}{T_0} \left( -\frac{mc^2}{2} \sqrt{\frac{\alpha a}{m}} \int \frac{\cos^2 \xi}{1 - e\cos \xi} d\xi + \frac{m}{2} \sqrt{\frac{\alpha a}{m}} \int \frac{d\xi}{1 - e\cos \xi} \right)$$

$$= \frac{1}{T_0} \left( \frac{m}{2} \sqrt{\frac{\alpha a}{m}} \int \frac{-e^2 \cos^2 \xi + 1}{1 - e\cos \xi} d\xi \right)$$

$$= \frac{1}{T_0} \left( \frac{m}{2} \sqrt{\frac{\alpha a}{m}} \int_0^{2\pi} (1 + e\cos \xi) d\xi \right)$$

$$\langle T \rangle = \frac{\alpha}{2a}$$

3. Time evolution in the Keplerian potential: Consider a particle of reduced mass  $m$  that is moving on a hyperbolic trajectory in the central potential  $U(r) = -\alpha/r$ , where  $\alpha > 0$ . Show that the time evolution of the trajectory can be parametrically expressed as follows:

$$r = a (e \cosh \xi - 1), \quad t = \sqrt{m a^3 / \alpha} (e \sinh \xi - \xi)$$

or, equivalently, as

$$x = a (e - \cosh \xi), \quad y = a \sqrt{e^2 - 1} \sinh \xi,$$

where the quantity  $a$  and the eccentricity of the orbit  $e$  are given in terms of the energy  $E > 0$  and the angular momentum  $\ell$  by the relations

$$a = \frac{\alpha}{2E}, \quad e = \sqrt{1 + \frac{2E\ell^2}{m\alpha^2}},$$

while  $-\infty < \xi < \infty$ .

(3)

$$t = \sqrt{\frac{m}{2}} \int \frac{dr}{\sqrt{E + \frac{\alpha}{r} - \frac{L^2}{2mr^2}}}$$

$$a = \frac{\alpha}{2E}$$

$$t = \sqrt{\frac{m}{2E}} \int \frac{dr}{\sqrt{1 + \frac{2a}{r} - \frac{L^2}{2mEr^2}}}$$

$$p = \frac{L^2}{m\alpha}$$

$$e^2 - 1 = \frac{2EL^2}{m\alpha^2} \Rightarrow$$

$$\frac{\alpha^2}{hE^2} (e^2 - 1) = \frac{L^2}{2mE}$$

$$a^2 (e^2 - 1) = \frac{L^2}{2mE}$$

$$t = \sqrt{\frac{m}{2E}} \int \frac{dr}{\sqrt{1 + \frac{2a}{r} - \frac{a^2(e^2-1)}{r^2}}}$$

$$t = \sqrt{\frac{m}{2E}} \int \frac{r dr}{\sqrt{r^2 + 2ar - a^2(e^2-1)}}$$

$$r = a(e \cosh \xi - 1)$$

$$r^2 = a^2(e^2 \cosh^2 \xi + 1 - 2e \cosh \xi)$$

$$\begin{aligned} r^2 + 2ar &= a^2(e^2 \cosh^2 \xi + 1 - 2e \cosh \xi + 2e \cosh \xi - 2) \\ &= a^2(e^2 \cosh^2 \xi - 1) \\ -a^2 e^2 + a^2 &= a^2 e^2 \sinh^2 \xi \end{aligned}$$

$$t = \sqrt{\frac{m}{2E}} \int a(e \cosh \xi - 1) d\xi$$

$$t = a \sqrt{\frac{m}{2E}} (e \sinh \xi - \xi)$$

$$\therefore \boxed{\begin{aligned} r &= a(e \cosh \xi - 1) \\ t &= \sqrt{\frac{ma^3}{2E}} (e \sinh \xi - \xi) \end{aligned}}$$

$$b) \quad x = r \cos \phi \quad y = r \sin \phi$$

$$r = \frac{p}{1 + \frac{ex}{r}} \Rightarrow r + ex = p$$

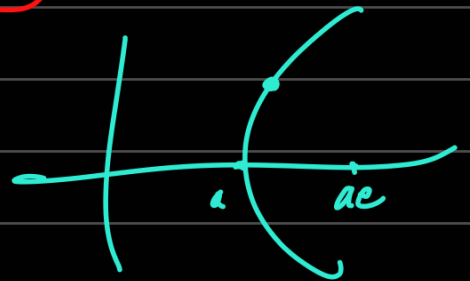
$$\Rightarrow ex = p - r$$

$$ex = a(e^2 - 1) - (ae \cosh \xi - a)$$

$$ex = ae^2 - a - ae \cosh \xi + a$$

$$x = ae - a \cosh \xi$$

$$\underline{x = a(e - \cosh \xi)}$$



*(Study parabola case)*

$$\begin{aligned}
 y &= \sqrt{r^2 - x^2} = \sqrt{a^2(1 - e \cosh \xi)^2 - a^2(e - \cosh \xi)^2} \\
 &= a \sqrt{1 + e^2 \cosh^2 \xi - 2e \cosh \xi - e^2 + 2e \cosh \xi - \cosh^2 \xi} \\
 &= a \sqrt{e^2 \sinh^2 \xi - \sinh^2 \xi} \\
 \underline{y} &= \underline{a \sqrt{e^2 - 1} \sinh \xi}
 \end{aligned}$$

4. Laplace-Runge-Lenz vector: Recall that, for a particle with  $s$  degrees of freedom, we require  $2s - 1$  constants of motion in order to arrive at a unique trajectory for the particle. According to this argument, for the Kepler problem, we would then need five integrals of motion to obtain the solution. We had expressed the solution in terms of the energy  $E$  of the system and the amplitude of the angular momentum vector  $\mathbf{L}$ , both of which were conserved. However, these quantities, viz. the energy  $E$  and the three components of the angular momentum vector  $\mathbf{L}$ , only add up to four constants of motion. Evidently, it will be interesting to examine if we can identify the fifth integral of motion associated with the system.

- (a) Show that, for a particle moving in the Keplerian central potential, i.e.  $U(r) = -\alpha/r$  with  $\alpha > 0$ , the following vector is an integral of motion:

$$\mathbf{A} = m \mathbf{v} \times \mathbf{L} - \frac{m \alpha \mathbf{r}}{r}.$$

Note: The conserved vector  $\mathbf{A}$  is known as the Laplace-Runge-Lenz vector.

- (b) Show that the vector  $\mathbf{A}$  lies in the plane of the orbit.  
 (c) Indicate the amplitude and the direction of  $\mathbf{A}$  associated with a planet as it moves in an elliptical orbit around the Sun.

Hint: Determine the amplitude and the direction of  $\mathbf{A}$  at, say, the perihelion and the aphelion.

- (d) If  $E$ ,  $\mathbf{L}$  and  $\mathbf{A}$  are all constants, then, we seem to have seven integrals of motion instead of the required five to arrive at a unique solution! How does seven reduce to five?

Hint: Examine if there exist any relations between  $\mathbf{A}$  and  $\mathbf{L}$  and/or  $E$ .

$$a) \quad \bar{\mathbf{A}} = m \bar{\mathbf{v}} \times \bar{\mathbf{L}} - \frac{m \alpha \bar{\mathbf{r}}}{r}$$

$$\frac{d\bar{\mathbf{A}}}{dt} = -\frac{\alpha}{r^2} \hat{\mathbf{r}} \times \bar{\mathbf{L}} - m \alpha \frac{d\hat{\mathbf{r}}}{dt}$$

$$\frac{d\bar{\mathbf{A}}}{dt} = \frac{\alpha L}{r^2} \hat{\boldsymbol{\phi}} - \frac{\alpha L}{r^2} \hat{\boldsymbol{\phi}} = \boxed{0}$$

$$\begin{aligned}
 \bar{\mathbf{L}} &= L \hat{\mathbf{z}} \\
 \frac{d\hat{\mathbf{r}}}{dt} &= \frac{\partial \hat{\mathbf{r}}}{\partial \phi} \dot{\phi} = \hat{\boldsymbol{\phi}} \frac{L}{mr^2}
 \end{aligned}$$

$$b) \quad \bar{\mathbf{A}} \cdot \bar{\mathbf{L}} = 0 \Rightarrow \underline{\bar{\mathbf{A}} \text{ is in plane}}$$

c) At perihelion,  $\vec{L} = L \hat{z}$ ,  $\vec{r} = \frac{L}{mr} \hat{y}$ ,  $\hat{r} = \hat{x}$   $r = a(1-e)$

$$\vec{A} = m\vec{v} \times \vec{L} - m\alpha \hat{r}$$

$$\vec{A} = \frac{L^2}{r} \hat{x} - m\alpha \hat{x}$$

$$\vec{A} \cdot \vec{A} = \left( \frac{L^2}{r} - m\alpha \right)^2 \hat{x} \cdot \hat{x} \quad r = \frac{L^2}{m\alpha(1+e)}$$

$$\vec{A} = m\alpha e \hat{x}$$

$$\Rightarrow |\vec{A}| = m\alpha e \quad \hat{A} = \hat{x}$$

d)  $\vec{A} \cdot \vec{L} = 0$  and  $|\vec{A}| = m\alpha \sqrt{1 + \frac{2EL^2}{mc^2}}$

Two independent constraints on  $\vec{A} \Rightarrow 7-2=5$  independent conserved quantities.

5. Passing through the centre: Consider a particle that describes a circular orbit under the influence of an attractive central force directed towards a point on the circle.

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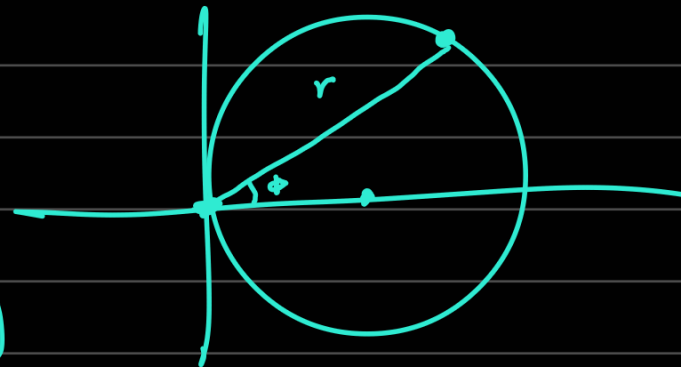
- (a) Show that the force varies as the inverse-fifth power of the distance.  
 (b) Also, show that for the orbit described the total energy of the particle is zero.  
 (c) Moreover, obtain the period of the motion.

$$r = 2R \cos \phi$$

$$E = \frac{1}{2} m \left( \frac{dr}{dt} \right)^2 \dot{\phi}^2 + \frac{L^2}{2mr^2} + V(r)$$

$$L = mr^2 \dot{\phi}$$

$$E = \left( \frac{1}{r^2} \frac{dr}{d\phi} \right)^2 \frac{L^2}{2m} + \frac{L^2}{2mr^2} + V(r)$$



$$u = \frac{1}{r} \quad \rightarrow \quad \frac{du}{d\phi} = -\frac{1}{r^2} \frac{dr}{d\phi}$$

$$E = \left( \frac{du}{d\phi} \right)^2 \frac{L^2}{2m} + \frac{L^2}{2m} u^2 + V\left(\frac{1}{u}\right)$$

$$0 = \left( \frac{du}{d\phi} \right) \frac{d^2 u}{d\phi^2} \frac{L^2}{m} + \frac{L^2}{m} u \left( \frac{du}{d\phi} \right) + \frac{dV}{du} \frac{du}{d\phi}$$

$$0 = \frac{d^2 u}{d\phi^2} \cdot \frac{L^2}{m} + \frac{L^2}{m} u + \frac{dV}{du}$$

$$u = \frac{\sec \phi}{2R} \quad \Rightarrow \quad \frac{du}{d\phi} = \frac{\sec \phi \tan \phi}{2R}$$

$$\frac{d^2 u}{d\phi^2} = \frac{\sec \phi \tan^2 \phi}{2R} + \frac{\sec^3 \phi}{2R}$$

$$0 = \frac{\sec \phi \tan^2 \phi}{2R} + \frac{\sec^3 \phi}{2R} + \frac{\sec \phi}{2R} + \frac{m}{L^2} \frac{dV}{du}$$

$$r = 2R \cos \phi \quad \Rightarrow \quad \sec \phi = \frac{2R}{r}$$

$$0 = \frac{\sec \phi}{2R} \left( \tan^2 \phi + \sec^2 \phi + 1 \right) - \frac{mr^2}{L^2} \frac{dv}{dr}$$

$$0 = \frac{\sec^3 \phi}{R} - \frac{mr^2}{L^2} \frac{dv}{dr}$$

$$0 = \frac{8R^2}{r^3} - \frac{mr^2}{L^2} \frac{dv}{dr} \Rightarrow -\frac{dv}{dr} = -\frac{8R^2 L^2}{mr^5}$$

$$\therefore \boxed{\vec{F}(r) = -\frac{8R^2 L^2}{mr^5} \hat{r}} \Rightarrow v(r) = -\frac{2R^2 L^2}{mr^4}$$

$$b) E = \left( \frac{1}{r^2} \frac{dr}{d\phi} \right)^2 \frac{L^2}{2m} + \frac{L^2}{2mr^2} - \frac{2R^2 L^2}{mr^4}$$

$$E = \frac{4R^2 \sin^2 \phi}{16R^4 \cos^4 \phi} \frac{L^2}{2m} + \frac{L^2}{8mR^2 \cos^2 \phi} - \frac{R^2 L^2}{8mR^4 \cos^4 \phi}$$

$$E = \frac{L^2}{8mR^2} \sec^2 \phi \tan^2 \phi + \frac{L^2 \sec^2 \phi}{8mR^2} - \frac{L^2 \sec^4 \phi}{8mR^2}$$

$$E = \frac{L^2 \sec^2 \phi}{8mR^2} \left( \tan^2 \phi + 1 - \sec^2 \phi \right) = \boxed{0}$$

$$c) \frac{T}{2} = \sqrt{\frac{m}{2}} \int_0^{2R} \frac{dr}{\sqrt{\frac{2R^2 L^2}{mr^4} - \frac{L^2}{2mr^2}}}$$

$$T = \sqrt{2m} \int_0^{2R} \frac{r^2 dr}{\sqrt{\frac{2R^2 L^2}{m} - \frac{L^2 r^2}{2m}}} = \sqrt{2m} \int_0^{2R} \frac{r^2 dr}{\frac{L}{\sqrt{m}} \sqrt{2R^2 - \frac{r^2}{2}}}$$

$$T = \frac{\sqrt{2m}}{L} \int_0^{2R} \frac{r^2 dr}{\sqrt{4R^2 - r^2}}$$

$$r = 2R \cos \phi$$

$$dr = -2R \sin \phi d\phi$$

$$T = \frac{m\sqrt{2}}{L} \int_0^{\frac{\pi}{2}} \frac{8R^3 \cos^2 \phi \sin \phi d\phi}{2R \sin \phi}$$



$$\begin{aligned}
 T &= \frac{m\sqrt{2}}{L} \int_0^{\frac{\pi}{2}} 4R^2 \cos^2\phi \, d\phi = \frac{2mR^2\sqrt{2}}{L} \int_0^{\frac{\pi}{2}} (1 + \cos 2\phi) \, d\phi \\
 &= \frac{2mR^2\sqrt{2}}{L} \left( \phi + \frac{\sin 2\phi}{2} \right) \Big|_0^{\frac{\pi}{2}} \\
 &= \frac{2\sqrt{2}mR^2}{L} \left( \frac{\pi}{2} \right) = \boxed{\frac{\pi\sqrt{2}mR^2}{L}}
 \end{aligned}$$